

COS30018 – Intelligent Systems

Week 9 tutorial

This week we will cover the following items:

- Reinforcement learning (RL) libraries/frameworks introduction (gymnasium and gym)

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Gymnasium (<https://github.com/farama-foundation/gymnasium>) is an open-source Python library used to develop, test, and compare reinforcement learning (RL) algorithms. Maintained by the **Farama Foundation**, it is the official, updated successor to the widely popular **OpenAI Gym** toolkit (<https://github.com/openai/gym>).

In reinforcement learning, an AI "agent" learns to make decisions by performing actions within a simulated "environment" to maximize a numerical reward. Gymnasium provides the standard "contract" or API that bridges your learning algorithm with these simulations.

Core Concepts and Features:

- **Standardized API:** It provides a unified interface so that an algorithm written for one task can easily run on another with minimal changes.
- **The Interaction Loop:** The core framework relies on simple methods:
 - `gym.make()`: Creates an instance of a specific environment `env` (after `import gymnasium as gym`)
 - `env.reset()`: Starts a new episode and returns the initial observation.
 - `env.step(action)`: Sends an action to the environment and returns the next observation, the reward, and signals indicating if the episode ended.
- **Separation of Signals:** Unlike older toolkits, Gymnasium cleanly separates episode completion into terminated (the task naturally ended, like failing a game) and truncated (the episode hit an artificial time limit).
- **Pre-Built Environments:** It comes bundled with standard benchmark suites, including:
 - **Classic Control:** Mathematical and physics tasks like Taxi, CartPole and MountainCar.
 - **Atari:** Classic retro video games.
 - **MuJoCo:** Physics-based robotic simulations.
 - **Box2D and Toy Text:** Gridworlds, blackjack, and car racing tasks.

Why Gymnasium Matters?

Before OpenAI Gym and Gymnasium, researchers built custom, non-standardized environments, making it nearly impossible to fairly compare different AI algorithms or reproduce research results. Gymnasium solved this by providing a universal baseline standard used across the global AI research community.

In this lab, we have taken an extract from the excellent course Practical_RL by Yandex School of Data Analysis (Week 3 - Model-free). You'll be provided with a notebook **Lab_Week 9.ipynb**.

Task 1: (in class) Some pieces of code within the notebook have been removed and your task is to complete the notebook with the correct pieces of code. You can do this in **Google Colab** to get an understanding of how different RL algorithms can be implemented and tested in **gymnasium**.

Task 2: (at home) Install the required Python libraries on your local machine so that you can run and test your RL algorithms without having to use Google Colab.